Orthogonality of Bessel functions

$$\int_0^b x\, J_p(k_m x)\, J_p(k_n x)\, dx = 0 \qquad m \neq n$$

proof: $x^2 J_p''(x) + x J_p'(x) + (x^2 - p^2) J_p(x) = 0$

$x \longrightarrow kx \qquad \frac{\partial}{\partial x} = \frac{1}{k}\frac{\partial}{\partial x}$

$$\Rightarrow \quad x^2 J_p''(kx) + x J_p'(kx) + (k^2x^2 - p^2) J_p(kx) = 0$$

$$\Rightarrow \quad x \frac{d}{dx}\left[x y'\right] + (k^2x^2 - p^2) J_p(kx) = 0$$

Thus, $J_p(k_m x)$ $J_p(k_n x)$ satisfy the following relations

$$x \frac{d}{dx}\left[x \frac{d}{dx} J_p(k_m x)\right] + (k_m^2 x^2 - p^2) J_p(k_m x) = 0$$

$$x \frac{d}{dx}\left[x \frac{d}{dx} J_p(k_n x)\right] + (k_n^2 x^2 - p^2) J_p(k_n x) = 0$$

$\forall \quad x^{-1} J_p(k_{r,m} x)$

$$J_p(k_n x) \frac{d}{dx}\left[x \frac{d}{dx} J_p(k_m x)\right]$$

$$J_\rho(k_n x)\frac{d}{dx}\left[x\frac{d}{dx}J_\rho(k_m x)\right] + \left(k_m^2 x - \frac{\rho^2}{x}\right) J_\rho(k_m x) J_\rho(k_n x) = 0$$

$$\Rightarrow \quad \left(k_m^2 - k_n^2\right)\int_0^b x\, J_\rho(k_m x) J_\rho(k_n x)$$

$$= \int_0^b J_\rho(k_m x)\frac{d}{dx}\left[x\frac{d}{dx}J_\rho(k_n x)\right]dx$$

$$- \int_0^b J_\rho(k_n x)\frac{d}{dx}\left[x\frac{d}{dx}J_\rho(k_m x)\right]dx$$

$u = J_\rho(k_m x)$ $\qquad$ $v = [x J_\rho'(kx)]$

$du = J_\rho'(k_m x)$ $\qquad$ $dv = [x J_\rho'(k_n)]'$

$$J_\rho(k_m x)\left[x J_\rho'(k_n)\right]\Big|_0^b - \int x J_\rho'(k_m x) J_\rho'(k_n x)$$

Since $J_\rho(k_m b) = 0$

$$\Rightarrow \quad -x\int J_\rho'(k_m x) J_\rho'(k_n x)\, dx$$

$$\Rightarrow \left(k_m^2 - k_n^2\right)\int_0^b x J_\rho(k_m x) J_\rho(k_n x)\, dx = 0$$

$$(k_m^2 - k_n^2) \int_0^b x\, J_p(k_m x)\, J_p(k_n x)\, dx$$

$$= x J_p(k_m x)\, J_p'(k_n x) - x J_p(k_n x) J_p'(k_m x) \Big|_0^b$$

$\lim k_m \longrightarrow k_n$

$$\Rightarrow \lim_{k_m \to k_n} \frac{x J_p(k_m x) J_p'(k_n x) - x J_p(k_n x) J_p'(k_m x)}{k_m^2 - k_2^2}$$

$$= \frac{x}{2k_m} \left[\frac{d}{dk_m} J_p(k_m x)\, J_p'(k_n x) - J_p(k_n x) \frac{d}{dk_m} J_p'(k_m x) \right] \Big|_0^b$$

since $\frac{d}{dx} J_p(kx) = \frac{p}{x} J_p(kx) - k J_{p+1}(kx)$

$x \longleftrightarrow k$

$$\Rightarrow \frac{d}{dk} J_p(kx) = \frac{p}{k} J_p(kx) - x J_{p+1}(kx)$$

$$= \frac{x}{2k} \left[\frac{d}{dk} J_p \frac{d}{dx} J_p - J_p \frac{d}{dk} \frac{d}{dx} J_p \right] \Big|_0^b$$

$$= \frac{x}{2k}\left[\frac{d}{dk}J_p \; \frac{d}{dx}J_p\right]\Big|_b$$

$$\left(\frac{p}{k}J_p - xJ_{p+1}\right)\left(\frac{p}{x}J_p - kJ_{p+1}\right)\Big|_b$$

$$\frac{p^2}{kx}J_p^2 - 2pJ_pJ_{p+1} + kxJ_{p+1}^2\Big|_b$$

$$\Rightarrow \boxed{\frac{b^2}{2}J_{p+1}^2} \quad \#\#$$